

Affine extension from equal-radius closed balls

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_closedBall`

theorem

Recovers the centers, matches the interiors, and then extends across the boundary.

Statement

Let E, F be real normed spaces, $c \in E$, $d \in F$, and $r > 0$. Write $\overline{B}(c, r) = \{x : \|x - c\| \leq r\}$. Every bijective isometry $f : \overline{B}(c, r) \rightarrow \overline{B}(d, r)$ extends uniquely to a surjective real affine isometry $A : E \rightarrow F$. The extension satisfies $A(x) = f(x)$ for every point of the source closed ball.

Assumptions

The source and target closed balls have the same positive radius r . No completeness or finite dimensionality is assumed.

Conclusion

There is one and only one ambient affine isometry agreeing with the given f on the entire closed ball, including its boundary.

Proof route

Use bounded reflection symmetry to send the centers to each other. The resulting distance-to-center identity verifies the interior equivalence required by the convex extension theorem.

▼ Proof steps

1. The centers belong to their balls since $r > 0$, and the source ball is bounded. Reflection through c preserves its ball because $\|(2c - x) - c\| = \|x - c\|$, and similarly for d ; see Center reflection preserves a closed ball. Apply Center transport by point reflection to these two balls and their centers to obtain $f(c) = d$.

2. For every $x \in \overline{B}(c, r)$,

$$\|f(x) - d\| = \|f(x) - f(c)\| = \|x - c\|.$$

For a positive radius, the ambient interior of each closed ball is its corresponding open ball. Therefore

$$\begin{aligned} x \in \text{int } \overline{B}(c, r) &\iff \|x - c\| < r \\ &\iff \|f(x) - d\| < r \\ &\iff f(x) \in \text{int } \overline{B}(d, r). \end{aligned}$$

The common radius is used in the middle equivalence.

3. The source closed ball is convex and its interior contains c . These facts and Step 2 verify Extension through a dense convex interior for the same f . That theorem returns the unique ambient extension with agreement on the whole source set, so it includes the boundary points as well.

Main citations

- Exact theorem and proof
- Center transport by point reflection
- Center reflection preserves a closed ball
- Extension through a dense convex interior

Lean source signature (exact)

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theorem existsUnique_affineExtension_closedBall
  {c : E} {d : F} {r : ℝ} (f : closedBall c r ≈i closedBall d r)
  (hr : 0 < r) :
  ∃! A : E ≈a.i[ℝ] F,
    ∀ x : closedBall c r,
      A (x : E) = ((f x : closedBall d r) : F)
```

In the source	Mathematical meaning
$\{c : E\} \{d : F\} \{r : \mathbb{R}\}$	The two centers and one common real radius in real normed spaces E, F .
$(f : \text{closedBall } c \ r \approx_i \text{closedBall } d \ r)$	The bijective isometry of the two closed balls with that same radius.
$(hr : 0 < r)$	The strict positivity $r > 0$.
$\exists! A : E \approx^{a.i}[\mathbb{R}] F$	Existence and uniqueness of a surjective real affine isometry $A : E \rightarrow F$ with the stated agreement.
$\forall x : \text{closedBall } c \ r,$	The agreement is required for every x satisfying $\ x - c\ \leq r$, including boundary points.
$A (x : E) = ((f x : \text{closedBall } d \ r) : F)$	The entire conclusion is $A(x) = f(x)$ at all those points, with the same A .

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_closedBall

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